

5. In 2010, scientists in China carried out an investigation to compare the growth of seeds on Earth and on the moon.

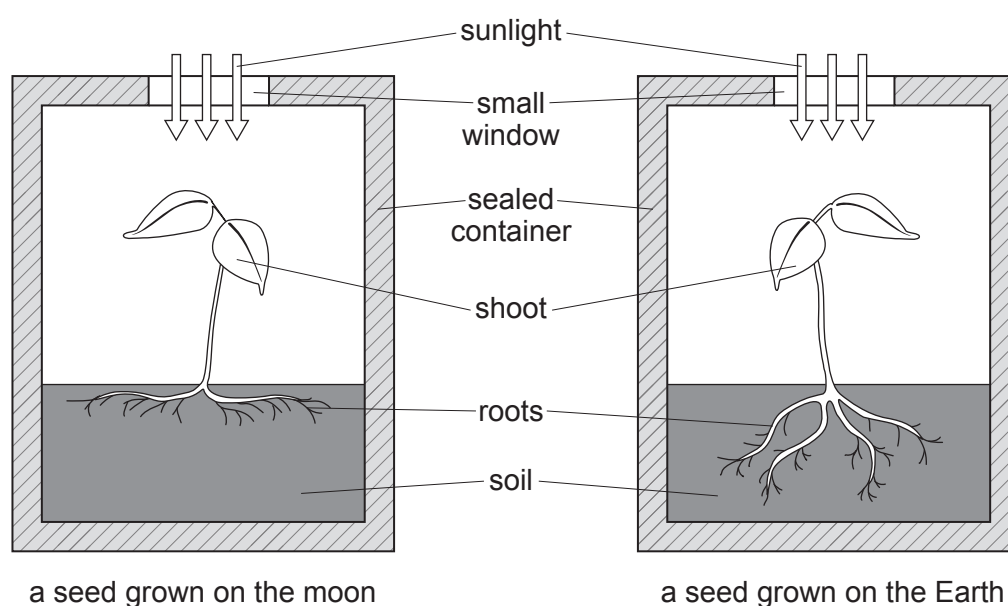
They sent seeds in a sealed container on board a rocket to the moon. The sealed container contained all the necessary conditions to allow the seeds to grow. The sealed container had a small window at the top, which allowed sunlight to reach the growing seeds. The container was at the same atmospheric pressure as the Earth.

An identical sealed container which contained identical seeds was kept on Earth. The scientists observed the seeds growing in both containers.

Images 5.1A and **5.1B** show the results of the investigations.

Image 5.1A

Image 5.1B



- (a) State the name of:

- (i) the growth response shown by the **shoots** in **Image 5.1A** and **Image 5.1B**. [1]

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- (ii) the hormone that controls growth responses in plants. [1]

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- (b) (i) Describe **one** difference in the growth of the roots in **Images 5.1A** and **5.1B**. [1]

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- (ii) Suggest **one** reason for the difference between the growth of seeds grown on the moon and those grown on Earth. [1]

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- (c) State the term used for carrying out an identical investigation on Earth. Suggest why this is important. [2]

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